

Numerical Relativity: when and why? and how?

Introduction to Numerical Relativity

47th Heidelberg Physics Graduate Days

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When *not* to try Numerical Relativity

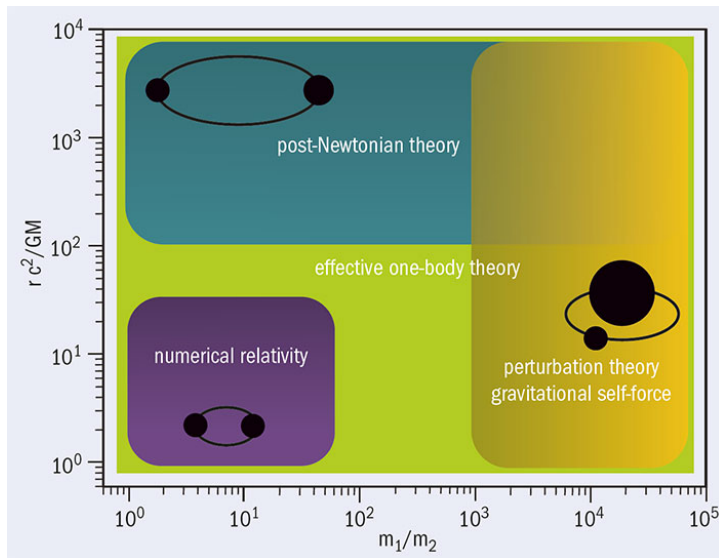
In GR we seek solutions to the field equations,

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R = 8\pi T_{ab}.$$

Describe how curvature of spacetime evolves and is related to matter distribution. Non-linear PDEs. Find solutions? Different physics:

- Is the desired solution time independent, or with a lot of symmetry? There might be an exact solution. Read a book.
- Is the desired solution close to an exact solution that we already know? Perturbation theory. Allow for bigger perturbations by going to second-order.
- Compact objects well-separated and moving slowly? Use Newtonian or post-Newtonian expansion to construct accurate approximate solutions.

Regimes of the relativistic two-body problem



Credit:
Buonanno
and
Sathyaprakash,
arXiv:1410.7832

When to try Numerical Relativity

On the other hand:

- Are the fields highly dynamical? Are non-linear terms expected to dominate the behavior? Is there no symmetry?
→ Use Numerical Relativity; numerical methods to find approximate solutions that converge to the continuum solution as you use more computational resources.

Ultimately, think of NR as just one of the ways to study gravity; the only one that can reach some solutions. It is really a part of computational physics, although for historical reasons, regrettably parts of NR have developed independently.

A brief history of Numerical Relativity

From Wikipedia NR page and Smarr:

- Late 1950s: ADM decomposition.
- 1960s. First numerics, including Smarr [nice talk: <http://online.itp.ucsb.edu/online/numrel00/smarr/>].
- 1970s. Axisymmetric spacetimes. York 3+1 decomposition.
- 1980s. Numerical Relativity becomes a field.
- 1990s. Critical Phenomena. 3D work becomes feasible. Grand challenge.
- 2000s. Compact binaries!
- 2010s. Accuracy. Template banks. GW detection!
- 2020s. ???

Overview for doing Numerical Relativity successfully

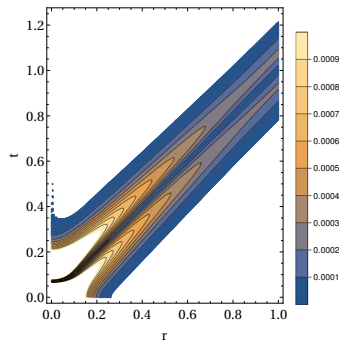
This and the next 7 slides contain a recipe of a sensible way to do NR.
The seven points are:

- ➊ Identify physical problem.
- ➋ Formulation.
- ➌ PDEs analysis.
- ➍ Select numerical method.
- ➎ Implementation. Extensive error testing.
- ➏ Evaluate errors.
- ➐ Physical interpretation.

A recipe for Numerical Relativity 1

1. Identify physical problem to solve. Obviously this should not belong to the class identified in the first slide. Examples include:

- Principle: Collapse of scalar or gravitational wave.
- GW astronomy: Compact binaries. Most important current problem.
- Astrophysics: Supernova core collapse.



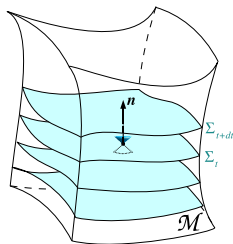
Gautam et. al: arXiv:gr-qc/2101.05038

Must understand limits described by other methods. Otherwise if something weird is happening in the simulation, you'll have *no idea*.

A recipe for Numerical Relativity 2

2. Formulation of the field equations. There is great freedom in writing down the field equations. Different methods:

- Constraints: conformal decomposition.
- Free-evolution.
- Constrained evolution.
- Characteristic formulation.



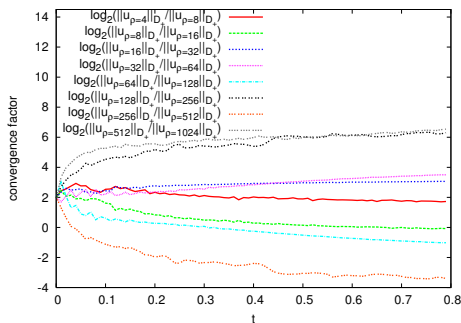
Brilliant notes: Gourgoulhon. [arXiv:gr-qc/0703035](https://arxiv.org/abs/gr-qc/0703035)

To minimize computational cost, we should also choose appropriate coordinates, symmetry reduction, and evolved quantities. Basically geometric consideration.

A recipe for Numerical Relativity 3

3. PDEs analysis of the formulation. The name says it all. Remember, you *must* adjust formulation if necessary. “necessary”?

- Well-posedness: Unique solution, depending continuously on given data.
- If not PDE unsatisfactory; no numerical approach can work. What happens?
- Sounds like GR is broken. Really a feature of gauge freedom.



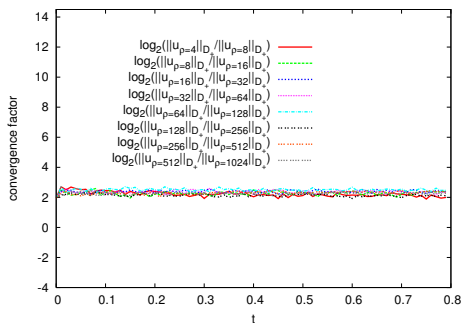
Hilditch & Cao: arXiv:1111.2177

Going beyond short-term existence becomes *very* mathematical. One place where NR meets mathematical relativity.

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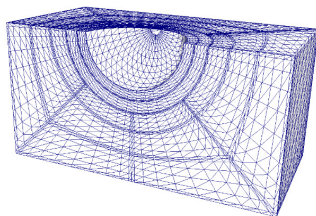
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A recipe for Numerical Relativity 4

4. Select numerical method(s). Examples include:

- Method of lines.
- Finite differencing.
- Pseudo-spectral methods.
- FMR, AMR.
- Summation by parts.
- Multipatch.
- High-Resolution-Shock-Capturing.
- Discontinuous Galerkin.



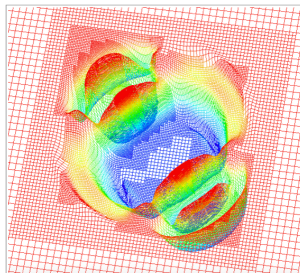
Szilágyi et.al: arXiv:0909.3557v1

Then use whatever methods of numerical analysis are available to see if approximation method can succeed. Often no guarantee.

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Clough et.al: arXiv:1503.03436

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A recipe for Numerical Relativity 5

5. Implementation of numerical method. Some considerations:

- Bad design decisions can hurt if you have to rewrite.
- Language, libraries, OSS. Reinventing bad if avoidable.
- Find code monkey. Make master and PhD students become one.
- Test *everything*. Document. Use weak field solutions.
- Optimization. MPI, Open MP, GPU computing. Optimize. Make code fast.



All of this is hard and the skill is *very* important both in NR and elsewhere.

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A recipe for Numerical Relativity 6

6. Evaluate errors in the numerical work. Perform convergence tests. Solve problem at higher resolution, comparing difference between data. If errors improve with resolution, but not sufficient:

- Go back and run at a higher resolution. But may have to beg for more supercomputer time.
- Develop more accurate methods.
- Make do.

Numerical Relativists spend their lives doing this; *importance can not be overstated*. Without trustworthy error estimates data worthless. Numerical analysis vs. computational physics.

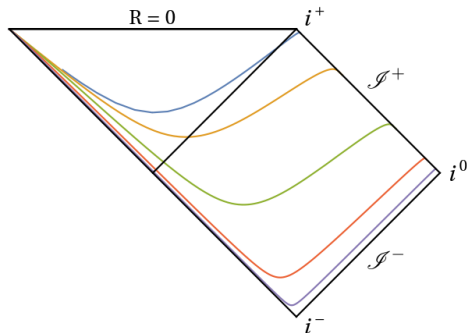
A recipe for Numerical Relativity 7

7. Physical interpretation of the results. NR meets mathematical relativity, geometry or astrophysics. Need to manipulate data; visualization, postprocessing. Examples:

- GW extraction.
- Event horizons.
- Conformal diagrams.
- Pretty pictures to impress students.

General comments:

- Real world means compromise.
- NR multi-disciplinary.
- This course: some ingredients of this recipe.



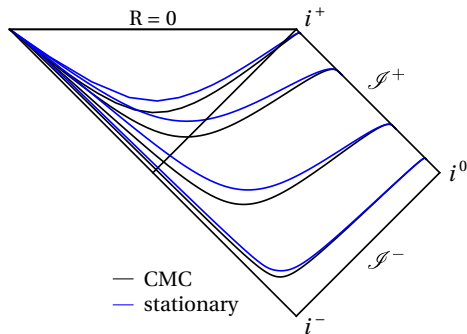
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Case study: critical phenomena in gravitational collapse

In the early 1990s [Matt Choptuik, PRL 70, 9-12 (1993)]. NR was fundamental in the discovery of previously unknown phenomena in gravitation.

- ① Physical problem: black-hole formation from collapsing matter.
- ② Formulation. Spherical symmetry. Constrained evolution.
- ③ PDEs analysis: Wave equation coupled to some ODEs.
- ④ Numerics: adaptive mesh-refinement.
- ⑤ Implementation.
- ⑥ Evaluate errors.
- ⑦ At threshold of BH formation, oscillations occur on ever smaller scales with same structure. Generates singularity *not* contained in an event horizon. Naked singularity!

Bibliography

- M.Alcubierre, *Introduction to 3+1 Numerical Relativity*, 2008
- T.W.Baumgarte and S.L.Shapiro, *Numerical Relativity: Starting from Scratch*, 2021
- T.W.Baumgarte and S.L.Shapiro, *Numerical Relativity: Solving Einstein's Equations on the Computer*, 2010
- E.Gourgoulhon, *3+1 Formalism in General Relativity: Bases of Numerical Relativity*, 2012
- M. Shibata, *Numerical Relativity – 100 Years of General Relativity: Volume 1*, 2015
- J.R.Wilson and G.J.Mathews, *Relativistic Numerical Hydrodynamics*, 2003

Further reading / specialised topics: Living Reviews in Relativity.

Extras for the course

Main and extra material in `https://fascinatingphysics.uibvirtual.es/numerical-relativity/`

- Mathematica notebooks with derivations that use the xAct package `http://xact.es/`
- Two small codes related to the course.
- List of reference textbooks.
- List of NR videos.